

ROTVAER, Norway

WMO: 011050

Lat: 68.368N

Lon: 15.943E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.3	-6.7	-13.5	1.2	-6.9	-12.0	1.3	-5.7	22.8	4.9	20.3	3.2	9.0	70	0.893

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	20.0	14.6	18.0	13.8	16.6	13.0	15.2	18.5	14.3	17.2	13.4	16.1	3.9	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.5	9.6	16.8	12.7	9.2	16.0	11.9	8.7	14.9	42.7	18.6	40.1	17.4	37.7	16.2	19.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	16.3	14.2	DB	-9.8	24.1	2.8	3.3	-11.8	26.5	-13.4	28.4	-15.0	30.3	-17.0	32.6	(4)
(5)			WB	-10.7	16.4	2.3	1.0	-12.3	17.1	-13.7	17.7	-15.0	18.2	-16.7	19.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-0.7	-1.1	0.0	2.8	6.8	9.9	13.1	13.0	10.5	6.1	3.3	1.3
	DBStd	5.74	3.30	3.41	2.73	2.30	2.99	2.71	2.70	2.23	2.23	2.69	2.82	2.87
	HDD10.0	1926	331	310	311	216	108	33	1	2	19	123	202	269
	HDD18.3	4700	589	543	569	466	358	253	164	164	234	378	452	528
	CDD10.0	271	0	0	0	0	9	30	98	96	36	3	0	0
	CDD18.3	4	0	0	0	0	0	1	3	1	0	0	0	0
	CDH23.3	10	0	0	0	0	1	1	7	1	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	7.0	7.9	7.8	7.6	6.6	6.5	6.4	6.2	6.1	6.2	7.0	7.3	7.9
Precipitation	PrecAvg	1542	161	147	149	91	73	65	85	106	134	190	163	172
	PrecMax	1949	267	248	230	163	133	126	161	248	279	380	303	314
	PrecMin	1319	53	74	28	32	24	22	27	46	61	50	49	83
	PrecStd	168	60	51	56	40	30	31	44	44	53	79	87	70
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	6.1	6.7	10.5	19.2	20.9	23.8	20.9	16.9	13.5	9.8	8.2
		MCWB	4.7	4.4	3.8	6.2	12.0	13.4	17.4	15.0	12.2	9.7	8.2	6.0
	2%	DB	6.1	5.4	5.5	8.5	14.5	17.4	20.7	18.7	15.6	12.0	8.8	7.3
		MCWB	4.4	3.9	3.5	5.3	9.6	11.6	15.3	14.3	11.4	9.0	6.9	5.6
	5%	DB	5.0	4.7	4.8	6.6	12.4	15.2	18.7	17.3	14.4	10.8	8.2	6.5
		MCWB	3.7	3.3	3.1	4.2	8.5	10.9	14.5	13.8	11.1	8.8	6.2	4.6
	10%	DB	3.9	3.7	4.1	5.5	10.5	13.6	17.0	16.2	13.2	9.7	7.3	5.4
		MCWB	2.4	2.3	2.7	3.6	7.3	10.3	13.5	13.2	10.8	7.9	5.3	3.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	5.2	4.7	6.7	12.4	13.9	17.9	16.1	12.9	10.4	8.5	7.0
		MCDB	6.3	5.8	5.7	10.0	19.7	19.8	22.0	18.7	15.3	11.9	9.2	7.5
	2%	WB	4.6	4.2	4.0	5.5	10.0	12.5	16.0	14.9	12.2	9.8	7.6	6.0
		MCDB	5.6	5.0	5.0	7.6	14.0	16.0	20.0	17.9	14.4	11.4	8.6	7.2
	5%	WB	3.6	3.4	3.3	4.7	8.6	11.5	14.8	14.1	11.6	8.9	6.5	4.7
		MCDB	4.9	4.5	4.6	6.1	11.9	14.4	18.0	16.7	13.6	10.7	7.9	6.3
	10%	WB	2.5	2.4	2.7	4.1	7.6	10.5	13.9	13.4	11.1	8.0	5.4	3.6
		MCDB	3.8	3.7	4.1	5.3	10.1	13.2	16.6	15.7	12.9	9.8	7.1	5.2
Mean Daily Temperature Range	5% DB	MDBR	2.9	3.0	3.1	3.3	4.0	3.8	4.2	3.7	3.0	2.5	2.8	2.9
		MCDBR	3.7	3.9	3.5	4.4	7.6	7.1	7.9	6.1	4.5	3.1	3.5	4.3
		MCWBR	3.3	3.6	2.9	2.7	3.4	3.3	3.4	2.8	2.3	2.4	3.1	4.0
	5% WB	MCDBR	3.6	3.8	3.6	3.8	6.9	6.1	7.4	5.5	3.6	2.8	3.5	4.3
		MCWBR	3.4	3.7	3.2	2.8	3.3	3.2	3.5	2.8	2.2	2.8	3.5	4.2
Clear-Sky Solar Irradiance	taub	0.372	0.272	0.310	0.331	0.340	0.341	0.346	0.336	0.319	0.283	0.352	N/A	
	taud	2.647	2.117	2.206	2.318	2.429	2.496	2.511	2.528	2.543	2.276	2.609	N/A	
	Ebn at Noon	114	502	711	815	852	862	844	813	732	495	97	0	
	Edh at Noon	13	46	77	93	93	90	86	76	58	39	11	0	
All-Sky Solar Radiation	RadAvg	0.03	0.32	1.44	3.19	4.60	4.87	4.44	3.22	1.70	0.58	0.07	0.00	
	RadStd	0.00	0.07	0.22	0.27	0.37	0.68	0.48	0.32	0.17	0.11	0.01	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page